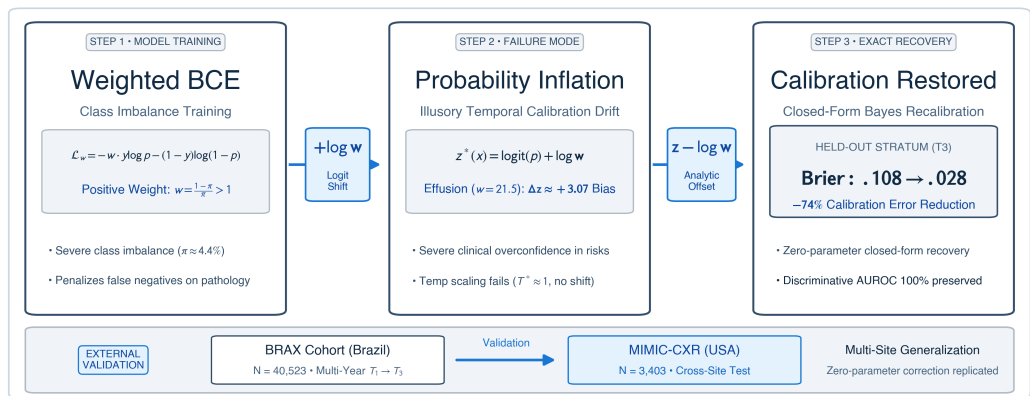


Graphical Abstract

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

Vishesh Panghal



Highlights

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

Vishesh Panghal

- Positive loss weighting induces an additive population logit offset of $\log w$.
- Finite neural network logits empirically cluster tightly around theoretical $\log w$.
- Temperature scaling cannot correct this intercept-type calibration bias.
- Analytic offset ($z - \log w$) cuts 74% of Brier error without retraining.
- Selective triage exposes trade-offs between calibrated risk and screening referral.

Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

Vishesh Panghal*

^aDepartment of Artificial Intelligence and Data Science, Poornima Institute of Engineering and Technology (PIET), Jaipur, 302022, Rajasthan, India

Abstract

Clinical deep learning models frequently exhibit calibration decay under deidentified-date cohort shift. Under extreme class imbalance, training with positive-weighted binary cross-entropy (BCE) is standard practice, but its downstream effect on probability calibration has been widely conflated with intrinsic feature drift. We resolve this conflation through a controlled 2×2 experimental design comprising two architectures and two loss objectives, with three independently trained random seeds per configuration (12 models total) on 40,523 chest radiographs from BRAX. Models are trained on an anchor partition and evaluated across later deidentified-date strata (T_1, T_2, T_3) with strict patient isolation. We show that positive loss weighting mathematically induces an additive population logit shift of $\log w$, which temperature scaling structurally cannot correct. In finite networks, empirical logit differences cluster tightly around theoretical $\log w$ (mean $\Delta z = 3.29 \pm 0.81$ vs. theoretical 3.07 for Pleural Effusion). On stratum T_3 , zero-parameter analytic offset ($z - \log w$) reduces total Brier score by 74% ($\Delta \text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]), substantially reducing intercept miscalibration from $\alpha = -2.96$ to -0.45 without retraining or validation data. On an external MIMIC-CXR-JPG cohort (3,403 radiographs, 289 patients), discrimination proved pathology-dependent (Pleural Effusion AUROC 0.688; Cardiomegaly 0.523), with severe baseline calibration shift ($\text{BSS} < 0$). Analytic offset yielded significant Brier reductions for Cardiomegaly and Pneumonia, but worsened

*Corresponding author.

Email address: pvt.panghal@gmail.com (Vishesh Panghal)

Edema, delineating boundaries of zero-parameter recalibration under cross-institutional shift. Finally, selective prediction simulations expose operational trade-offs between risk estimation and high-sensitivity screening triage.

Keywords: Chest radiography, Probability calibration, Class imbalance, Distribution shift, Selective prediction, Deep learning, Clinical decision support

1. Introduction

Deep learning models for automated chest radiograph interpretation have achieved high discriminatory performance across common thoracic abnormalities such as pleural effusion, cardiomegaly, and pneumonia [1, 2]. However, the transition from stationary retrospective validation to longitudinal clinical deployment introduces substantial distribution shifts. In operational healthcare environments, patient populations evolve, imaging protocols shift, scanner hardware is replaced, and demographic or seasonal patterns alter condition prevalence over time [3, 4, 5].

While discriminatory ranking (measured via AUROC) often exhibits gradual longitudinal attenuation under temporal shift, decision-critical applications rely fundamentally on *calibrated posterior probabilities*: $P(Y = 1 \mid X = x)$ [6, 7]. When predicted probabilities deviate from true empirical event rates, downstream operational decision cutoffs shift unpredictably, expected utility formulations degrade, and automated decision-support pipelines generate distorted risk rankings.

Recent literature has reported rapid “temporal calibration decay,” asserting that frozen deep neural networks suffer progressive miscalibration over deployment periods [8, 9]. Simultaneously, because biomedical datasets are characterized by extreme class imbalance (pathology prevalence often below 5%), researchers routinely train vision models using positive-weighted binary cross-entropy (BCE) to penalize missed positive cases [1, 10].

In this work, we present a computational investigation showing that these two phenomena are deeply entangled, and that the prevailing attribution of temporal calibration degradation primarily to intrinsic feature drift may be largely explained by uncorrected post-training calibration offsets under label distribution shifts [11, 12]. Under population-risk minimization, when models are trained with positive loss weighting $w > 1$, the Bayes-optimal logit is shifted by an additive constant $+\log w$ [13, 14]. In theoretical ma-

chine learning, Caplin, Martin, and Marx [14] formally established that class weighting is fundamentally incompatible with unadjusted probability calibration and derived an inverse machine learning formulation to recover true likelihoods, demonstrating this on a static chest radiograph pneumonia task. Standard post-hoc calibration methods—most notably temperature scaling [15]—only rescale the logit variance (z/T) and cannot shift the logit intercept. Consequently, weighted models continuously over-predict positive probabilities, producing severe Brier score and Expected Calibration Error (ECE) degradation that compounds over time as disease prevalence fluctuates.

1.1. Related Work

Probability Calibration in Deep Learning: Modern deep convolutional neural networks frequently exhibit over-confidence due to unconstrained cross-entropy minimization with over-parameterized architectures [15]. Post-hoc recalibration techniques such as temperature scaling, histogram binning, and isotonic regression have been widely evaluated [15, 16]. In medical decision models, calibration accuracy is paramount for valid risk stratification [6, 7]. However, standard methods were primarily derived for multi-class classification under balanced distributions, leaving binary imbalanced settings less formally explored.

Class Imbalance and Loss Function Design: Extreme class imbalance is pervasive in thoracic imaging. Common mitigation strategies include positive loss weighting [10, 1], focal loss [17], and re-sampling. In clinical prediction modeling, Piccininni et al. [18] formally demonstrated that correcting class imbalance through resampling inherently alters the effective sampling proportion and damages probability calibration, while showing that plug-in probability corrections can recover valid absolute risks while discrimination remains largely preserved. In loss-based weighting, King and Zeng [13] established in classical logistic regression that choice-based sample proportion adjustments induce an intercept shift of $\log w$. More recently, Caplin, Martin, and Marx [14] analyzed loss-weighted deep learning through an economic modeling lens, rigorously establishing that class weighting is mathematically incompatible with raw posterior probability calibration. They formulated an inverse machine learning framework to recover undistorted likelihoods from weighted models and demonstrated feasibility on a static CheXNet pneumonia detection task [1]. However, their work was confined to stationary single-dataset evaluations, leaving unexamined how loss-induced logit shifts behave under sequential temporal cohort shifts, whether standard

deep calibration methods (temperature scaling) fail under this distortion, how finite neural networks optimize under extreme class scarcity, and how post-hoc recalibration behaves under cross-institutional transfer. Modern clinical pipelines routinely deploy weighted architectures without correcting for this induced logit offset prior to post-hoc calibration.

Distribution Shift and Label Shift Adaptation: Temporal dataset shift encompasses changes in feature distributions $P(X)$, conditional distributions $P(Y | X)$, and prior label probabilities $P(Y)$ [3]. Davis et al. [12] demonstrated that calibration drift often precedes discriminatory loss in clinical models. Alexandari et al. [11] established that maximum likelihood with bias-corrected calibration provides robust adaptation under label shift. Decoupling algorithmic training offsets from genuine temporal shift is essential for diagnostic integrity.

Selective Prediction and Deferral Mechanisms: Selective classification allows models to abstain from predicting on ambiguous inputs, routing them to human experts [19, 20]. Fisch et al. [21] highlighted that selective classification performance is fundamentally bounded by probability calibration. Recently, Aperstein et al. [22] demonstrated multi-pathology chest radiograph selective prediction across PadChest, NIH ChestX-ray14, and MIMIC-CXR using entropy- and confidence-based rejection mechanisms evaluated via risk-coverage profiles and bootstrap inference. While such studies focus on developing abstention policies, the interaction between underlying loss-induced calibration distortion and downstream selective triage remains largely unexplored. In this work, rather than proposing a new abstention algorithm, we utilize selective prediction as an operational downstream lens to evaluate the real-world clinical consequences of calibration mechanisms, exposing how loss weighting and calibration alter referral efficiency and screening recall under workflow constraints.

1.2. Contributions

To disentangle algorithmic loss bias from distribution shift across cohort strata, this paper provides:

1. **A Controlled 2×2 Experimental Design:** We execute a controlled 2×2 experimental design comprising two architectures and two loss objectives (DenseNet-121 and ResNet-50 $\times$ standard BCE and positive-weighted BCE), with three independently trained random seeds per configuration (12 models total) trained on 25,123 chest radiographs

and evaluated on 8,302 held-out radiographs across three sequential deidentified-date strata (T_1, T_2, T_3) from the BRAX dataset [2].

2. Theoretical Derivation and Empirical Verification of Logit

Shift: Extending foundational observations of loss-induced probability distortion [13, 14] into longitudinal clinical deep learning, we derive that positive-weighted BCE introduces a constant population logit offset $\log w$. Across 16,604 evaluations spanning two convolutional backbones and diverse thoracic pathologies, we empirically demonstrate where finite neural networks tightly adhere to theoretical $\log w$ and delineate where extreme class scarcity ($w > 1000$) causes optimization saturation. Crucially, we prove that standard temperature scaling is structurally incapable of correcting this logit intercept displacement ($\alpha \approx -2.96$), whereas an exact zero-parameter analytic offset ($z - \log w$) or non-negative Platt scaling restores calibration parity with unweighted models (substantially reducing intercept bias to $\alpha = -0.45$ for DenseNet-121 and reducing total Brier error by 74%–76%).

3. A Leak-Free Out-of-Sample Calibration Protocol:

We implement a strict 5-fold patient-clustered cross-fitting protocol on T_1 , strictly preventing in-sample calibration leakage, and report out-of-sample generalization on T_2 and T_3 backed by 2,000-replicate patient-clustered bootstrap confidence intervals directly on individual seeds and the 3-seed ensemble.

4. Operational Selective Prediction and Workload Triage Analysis:

Rather than proposing an independent abstention algorithm, we utilize selective prediction as an operational downstream lens with explicit, transparent tracking of reader referral workloads, retained sensitivity ($TP/N_{\text{pos, ret}}$), and full-cohort system sensitivity ($1 - FN/N_{\text{pos, total}}$) under matched development thresholds, demonstrating that calibrated models optimize probability estimation while raw weighted models support recall-oriented screening triage.

2. Materials and Methods

2.1. Dataset and Cohort Stratification

We utilize the Brazilian Chest X-ray (BRAX) v1.1.0 benchmark [2]. BRAX v1.1.0 contains 40,967 radiographs. After excluding 444 radiographs from 89 patients whose examinations crossed partition boundaries, the analytic cohort comprised 40,523 radiographs acquired across multiple inpatient and

139 outpatient emergency facilities at Hospital Israelita Albert Einstein in São
 140 Paulo, Brazil. Radiographs are labeled with 14 thoracic findings extracted
 141 from Portuguese radiological reports using a validated natural language
 142 processing pipeline.

143 In compliance with HIPAA de-identification standards, direct patient
 144 identifiers and original timestamps are protected. We extract chronological
 145 study dates from the released 8-digit calendar integers (YYYYMMDD), spanning
 146 2008 through 2017. BRAX documentation states that the released dates
 147 are fictitious dates generated through a hashing procedure while retaining
 148 the original intervals between study acquisitions. Because synchronization
 149 of anonymized dates across different patients is not explicitly established,
 150 we interpret these partitions as sequential deidentified-date strata rather
 151 than real-world calendar periods. To maintain strict partition integrity and
 152 preclude longitudinal data leakage:

- 153 1. All studies are partitioned chronologically around a strict model freeze
 154 boundary: $T_{\text{freeze}} = 2015-07-01$.
- 155 2. All training studies occur before 2013-09-01 ($N = 25,123$ images from
 156 11,265 patients across 14,315 studies).
- 157 3. All validation studies occur between 2013-09-01 and 2015-06-30 ($N =$
 158 7,098 images from 3,281 patients across 4,012 studies).
- 159 4. All held-out test studies occur between 2015-07-01 and 2017-12-17
 160 ($N = 8,302$ images from 3,807 patients across 4,670 studies), partitioned
 161 into three consecutive deidentified-date strata: T_1 (2015-H2, $N = 1,879$),
 162 T_2 (2016, $N = 3,987$), and T_3 (2017, $N = 2,436$).
- 163 5. Strict patient isolation is enforced: $\text{Patients}_{\text{train}} \cap \text{Patients}_{\text{val}} \cap \text{Patients}_{\text{test}} =$
 164 $\emptyset$. Specifically, 444 radiographs from 89 patients whose examinations
 165 crossed partition boundaries were excluded to guarantee zero within-
 166 patient contamination, yielding the analytic cohort of 40,523 radio-
 167 graphs.
- 168 6. For longitudinal paired logit offset and multi-calibrator evaluation,
 169 all 8,302 held-out radiographs ($N = 1,879$ in T_1 , $N = 3,987$ in T_2 ,
 170 and $N = 2,436$ in T_3) are evaluated across both clinical architectures
 171 (DenseNet-121 and ResNet-50) with zero held-out exclusions (0% ex-
 172 clusion rate), yielding 16,604 paired ensemble evaluations ($8,302 \times 2$
 173 image-architecture pairs with 3 seed replications averaged per pair;
 174 49,812 total seed-level pairs) with complete longitudinal tracking.

Table 1: Cohort Demographics, Radiographic Projections, and Target Condition Prevalence across Stratified Cohorts

Cohort Stratum	Images (N)	Patients (N)	Studies (N)	Sex (% M / F)	Age (Mean $\pm$ SD)	Views (% PA / AP / Lat / Unk)	Pleural Effusion N (%)	Cardiomegaly N (%)	Pneumonia N (%)	Edema N (%)
Train (Pre-deployment)	25,123	11,265	14,315	49.3% / 50.7%	46.0 $\pm$ 24.3	34.0% / 13.5% / 0.0% / 27.2%	1,116 (4.44%)	2,398 (9.55%)	456 (1.82%)	23 (0.09%)
Val (Pre-deployment)	7,098	3,281	4,012	49.8% / 50.2%	46.2 $\pm$ 24.1	34.7% / 12.2% / 0.0% / 27.7%	295 (4.16%)	681 (9.59%)	128 (1.80%)	8 (0.11%)
Test T1 (2015)	1,879	913	1,039	47.8% / 52.2%	44.5 $\pm$ 23.9	34.4% / 10.2% / 0.0% / 29.0%	99 (5.27%)	147 (7.82%)	50 (2.66%)	4 (0.21%)
Test T2 (2016)	3,987	1,832	2,249	48.5% / 51.5%	46.1 $\pm$ 23.8	34.6% / 12.0% / 0.0% / 27.6%	177 (4.44%)	422 (10.58%)	79 (1.98%)	13 (0.33%)
Test T3 (2017)	2,436	1,164	1,382	50.9% / 49.1%	46.0 $\pm$ 24.6	34.4% / 12.4% / 0.0% / 28.1%	85 (3.49%)	258 (10.59%)	50 (2.05%)	0 (0.00%)
Test All (Pooled)	8,302	3,807	4,670	49.0% / 51.0%	45.7 $\pm$ 24.1	34.5% / 11.7% / 0.0% / 28.1%	361 (4.35%)	827 (9.96%)	179 (2.16%)	17 (0.20%)

Demographic, projection, and disease prevalence characteristics across all cohort strata are detailed in Table 1.

2.2. External Stress-Testing Cohort (MIMIC-CXR-JPG)

To stress-test whether the mathematical calibration mechanism survives extreme cross-institutional domain shift—rather than claiming turnkey diagnostic generalization—frozen BRAX-trained models were externally evaluated on all eligible frontal radiographs (AP and PA views) from the official patient-partitioned MIMIC-CXR-JPG test partition [23, 24]. The cohort comprises $N = 3,403$ frontal radiographs ($N = 3,041$ diagnostic studies across $N = 289$ unique patients; AP projections: 2,405 [70.67%], PA projections: 998 [29.33%]). In strict accordance with the primary evaluation protocol, ground-truth diagnostic labels were extracted using the official CheXpert labeler under the standard uncertainty-zero (U -zero) policy. Empirical condition prevalence across images was: Pleural Effusion 32.18% ($N = 1,095$ positive radiographs), Cardiomegaly 26.33% ($N = 896$), Edema 21.33% ($N = 726$), and Pneumonia 10.05% ($N = 342$). Because multiple radiographs and studies belong to the same patient, patient-clustered non-parametric bootstrap inference was used throughout external evaluation to account for repeated within-patient observations, avoiding artificial inflation of effective sample size. All 3,403 radiographs were physically verified on disk (100.00% completeness). No retraining or hyperparameter fine-tuning was performed, isolating out-of-distribution recalibration behavior from in-domain adaptation.

2.3. Architectures and Training Objectives

We evaluate two standard clinical vision architectures initialized from ImageNet pre-trained weights:

- **DenseNet-121** [25]: Employs dense feature concatenation, standard for thoracic pathology benchmarks [1].
- **ResNet-50** [26]: Employs residual skip connections as an independent structural comparator.

Each architecture is trained under two distinct loss objectives:

1. Unweighted Binary Cross-Entropy (Standard BCE):

$$\mathcal{L}_{\text{std}}(y, \hat{p}) = -[y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (1)$$

where $\hat{p} = \sigma(z) = 1/(1 + e^{-z})$ is the sigmoid probability from raw logit z .

2. Positive-Weighted Binary Cross-Entropy (Weighted BCE):

$$\mathcal{L}_{\text{pos}}(y, \hat{p}) = -[w \cdot y \log \hat{p} + (1 - y) \log(1 - \hat{p})] \quad (2)$$

where $w = (N_{\text{total}} - N_{\text{pos}})/N_{\text{pos}}$ represents the inverse positive prevalence in the training set (e.g., $w = 21.51$ for Pleural Effusion, $w = 9.48$ for Cardiomegaly, $w = 54.09$ for Pneumonia).

Models are trained with the AdamW optimizer (initial learning rate 10^{-4} , weight decay 10^{-2}) and a Cosine Annealing learning rate schedule over 15 epochs, with batch size 64 and Automatic Mixed Precision (AMP fp16) on 512×512 pixel input resolution. Pleural Effusion was pre-specified as the primary target pathology because of its adequate prevalence and stable evaluability across all longitudinal held-out strata ($N = 1,879$ in T_1 , $N = 3,987$ in T_2 , and $N = 2,436$ in T_3), serving as the primary benchmark for all multi-calibrator comparisons and selective triage simulations; Cardiomegaly, Pneumonia, and Edema served as secondary mechanistic validation targets spanning contrasting baseline prevalence and class-scarcity regimes. Model checkpoints were accordingly selected based on the epoch achieving the highest validation Pleural Effusion AUROC across the 15 epochs (occurring between epochs 4 and 15 across runs), without early stopping loop termination. Each configuration is executed across three random seeds ($S \in \{42, 123, 2026\}$), producing 12 fully independent models. For unified system-level benchmarking, a 3-seed probability ensemble is constructed for each architecture and loss objective by averaging predicted probabilities across independent training runs:

$$\bar{p}_{\text{raw}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x)), \quad \bar{p}_{\text{corrected}}(x) = \frac{1}{3} \sum_{s \in S} \sigma(z_s(x) - \log w) \quad (3)$$

All post-hoc calibration, paired analytic adjustments, and selective prediction policies are evaluated directly on both individual model seeds and the unified

232 probability ensemble, ensuring strict mathematical consistency across all
 233 reported metrics and tables. For each image x and architecture, the empirical
 234 logit offset $\Delta z(x)$ was computed from the ensemble representations as the
 235 difference between the mean logits of the weighted and unweighted models
 236 across the three seed-matched pairs:

$$\Delta z(x) = \bar{z}_w(x) - \bar{z}_u(x) = \frac{1}{3} \sum_{s \in S} (z_{w,s}(x) - z_{u,s}(x)) \quad (4)$$

237 yielding exactly 16,604 ensemble-paired evaluations per pathology (8,302
 238 images $\times$ 2 architectures). Evaluating all individual seed-matched pairs
 239 independently without averaging yields $8,302 \times 2 \times 3 = 49,812$ paired evalu-
 240 ations, which exhibit algebraically identical sample means due to linearity
 241 of expectation ($\Delta z = 3.29$ for Pleural Effusion, 3.83 for Pneumonia, 2.64 for
 242 Cardiomegaly, and 3.69 for Edema).

243 2.4. Mathematical Mechanics of Loss Weighting and Calibration

244 Under positive-weighted binary cross-entropy, the empirical risk minimizer
 245 seeks to minimize the expected loss:

$$\begin{aligned} \mathbb{E}_{Y|X}[\mathcal{L}_{\text{pos}}(Y, \hat{p})] &= -w \cdot P(Y = 1 | X) \log \hat{p} \\ &\quad - P(Y = 0 | X) \log(1 - \hat{p}) \end{aligned} \quad (5)$$

246 Setting the derivative with respect to logit z to zero:

$$\begin{aligned} \frac{\partial}{\partial z} \mathbb{E}[\mathcal{L}_{\text{pos}}] &= \hat{p} [wP(Y = 1|X) + P(Y = 0|X)] \\ &\quad - wP(Y = 1|X) = 0 \end{aligned} \quad (6)$$

247 Solving for the optimal predicted probability $\hat{p}^*$:

$$\hat{p}^* = \frac{w \cdot P(Y = 1 | X)}{w \cdot P(Y = 1 | X) + P(Y = 0 | X)} \quad (7)$$

248 Converting to log-odds:

$$z^*(X) = \log \left(\frac{\hat{p}^*}{1 - \hat{p}^*} \right) = \log \left(\frac{P(Y = 1 | X)}{P(Y = 0 | X)} \right) + \log w \quad (8)$$

249 Equation (8) derives that positive loss weighting shifts the model’s population
 250 Bayes-optimal logit by $+\log w$, connecting classical choice-based sampling

corrections [13] with the class-weight likelihood recovery formulation of Caplin et al. [14]. When $w \gg 1$, the network is mathematically incentivized to output large positive logits, systematically inflating predicted probabilities. We emphasize that Equation (8) describes the population-risk minimizer under infinite capacity; in finite neural networks trained with stochastic gradient descent, empirical logits fluctuate around this theoretical attractor.

2.4.1. Limitation of Temperature Scaling

Temperature scaling [15] adjusts probabilities via a single positive scalar T :

$$p_{\text{temp}} = \sigma\left(\frac{z}{T}\right) \quad (9)$$

To formally evaluate calibration structure, we examine the standard logistic calibration regression (Cox calibration) [6, 7]:

$$\text{logit}(P(Y = 1 \mid X)) = \alpha + \beta \cdot \text{logit}(\hat{p}) \quad (10)$$

where $\alpha = 0$ represents perfect calibration-in-the-large (calibration intercept), and $\beta = 1$ denotes ideal calibration slope. Because positive loss weighting induces an additive shift $z \approx z_{\text{true}} + \log w$, the uncorrected logit suffers a massive intercept discrepancy ($\alpha \approx -\log w \ll 0$). Temperature scaling multiplies the logit by $1/T$, adjusting the slope $\beta \rightarrow \beta \cdot T$, but is mathematically incapable of shifting the additive constant α . In low-prevalence clinical domains where negative cases vastly dominate, the negative log-likelihood optimizer is constrained by the conflicting pressures of slope and intercept, typically selecting $T^* \approx 1.0$ or providing limited, architecture-dependent variance shrinkage without eliminating the severe underlying intercept displacement.

2.4.2. Post-Hoc Recalibration Suite

To systematically rectify this mechanism, we evaluate four post-hoc recalibration functions:

1. **Temperature Scaling (T^*):** $p = \sigma(z/T^*)$, optimized via bounded Brent minimization on $[0.01, 50.0]$ with an explicit monotonicity constraint (if $\text{NLL}(T^*) > \text{NLL}(1.0) \implies T^* = 1.0$).
2. **Analytic Offset ($z - \log w$):** Directly subtracts the theoretical training logit shift:

$$p_{\text{analytic}} = \sigma(z - \log w) \quad (11)$$

Requires zero validation data and zero numerical optimization, resetting $\alpha \approx 0$ by construction while preserving model discrimination and slope.

- 282 3. **Fitted Offset** ($z + a^*$): Optimizes a single scalar intercept $a^* \in$
 283 $[-10.0, 10.0]$ via univariate Brent minimization on validation log-likelihood,
 284 holding the slope fixed at 1.0.
- 285 4. **Platt Scaling** ($a + bz$): Logistic regression on logits $p = \sigma(a + bz)$
 286 with a non-negative constraint on slope ($b \geq 0$) to prevent spurious
 287 rank reversal.

288 2.4.3. Algorithmic and Computational Complexity

289 The evaluated post-hoc calibrators exhibit distinct computational profiles:

- 290 • **Temperature Scaling:** Incurs $O(N_{\text{val}})$ scalar optimization over vali-
 291 dation logits and $O(1)$ inference per prediction.
- 292 • **Analytic Offset:** Incurs strictly $O(1)$ closed-form scalar subtraction
 293 ($z - \log w$), requiring zero validation storage, zero parameter fitting,
 294 and $O(1)$ inference.
- 295 • **Fitted Offset:** Incurs $O(N_{\text{val}})$ univariate Brent optimization on valida-
 296 tion log-likelihood and $O(1)$ inference.
- 297 • **Platt Scaling:** Incurs $O(k \cdot N_{\text{val}})$ iteratively reweighted least squares
 298 fitting (where k is the number of optimization steps) and $O(1)$ inference.

299 The analytic offset thus provides an operational advantage for zero-compute
 300 edge deployment, eliminating the requirement to retain or access historical
 301 validation datasets.

302 2.5. Cross-Fitting and Resampling Evaluation Protocol

303 To ensure zero optimistic in-sample bias:

- 304 • On deployment stratum T_1 (2015-H2), calibrator parameters are esti-
 305 mated using **5-fold patient-clustered cross-fitting**. Calibrators are
 306 fit on 4 folds and evaluated out-of-fold on the 5th fold, with all studies
 307 from any given patient assigned to a single fold.
- 308 • For evaluation on later held-out deidentified-date strata T_2 (2016) and
 309 T_3 (2017), the calibrators fit on T_1 are frozen and applied strictly
 310 out-of-sample.
- 311 • **Statistical Inference:** All 95% confidence intervals (CIs) are com-
 312 puted using $B = 2,000$ patient-clustered bootstrap resamples with
 313 replacement. For paired calibrator comparisons (e.g., Analytic Offset
 314 vs. Raw), differences are computed on identical bootstrap resamples.

2.6. Selective Prediction and Deferral Policy

In operational deployment, models need not provide an autonomous prediction for every image. An uncertain model can selectively defer ambiguous examinations to expert human readers [19, 20, 22]. Rather than developing novel rejection algorithms, our selective prediction formulation serves as an operational downstream analysis to evaluate how loss weighting and probability calibration impact clinical referral efficiency and automated error rates.

To ensure fair comparisons across calibrators operating on different probability scales, we determine an operating decision cutoff t^* for each method strictly on development stratum T_1 using Youden’s J index ($J = \text{Sensitivity} + \text{Specificity} - 1$), without using later held-out evaluation labels. For unweighted models, $t^* \approx 0.014$; for analytic offset calibrated models, $t^* \approx 0.022$; and for raw weighted models, $t^* \approx 0.33$. We define a threshold-relative uncertainty metric based on piecewise distance from the operational cutoff t^* :

$$u(x) = 1 - \frac{|\hat{p}(x) - t^*|}{\mathbb{I}(\hat{p}(x) \geq t^*)(1 - t^*) + \mathbb{I}(\hat{p}(x) < t^*)t^*} \in [0, 1] \quad (12)$$

where $u(x) \rightarrow 1$ indicates maximal ambiguity ($\hat{p}(x) \approx t^*$) and $u(x) \rightarrow 0$ indicates decisive confidence. Crucially, because t^* scales with baseline disease prevalence (e.g., $t^* \approx 0.022$ for analytic calibrated models), predictions below t^* are normalized by t^* . Consequently, low-probability false negatives ($\hat{p} \ll t^*$) yield low uncertainty and are retained in the automated pool rather than referred. At coverage level $C \in [0.5, 1.0]$, the automated system retains the fraction C of cases with lowest uncertainty ($N_{\text{auto}} = \lfloor C \cdot N \rfloor$) and refers the remaining fraction $1 - C$ ($N_{\text{ref}} = N - N_{\text{auto}}$) to radiologist review.

To prevent metric conflation and provide complete arithmetic transparency, we evaluate triage outcomes using distinct, explicit definitions:

- **Retained Positives ($N_{\text{pos, ret}}$):** The number of true positive cases remaining in the automated cohort after ambiguous examinations have been referred to human readers.
- **Automated True Positives (TP_{auto}):** True pathology cases correctly flagged as positive by the autonomous system ($TP_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 1)$).
- **Automated False Negatives (FN_{auto} , Missed Cases):** True pathology cases erroneously classified as negative by the autonomous system

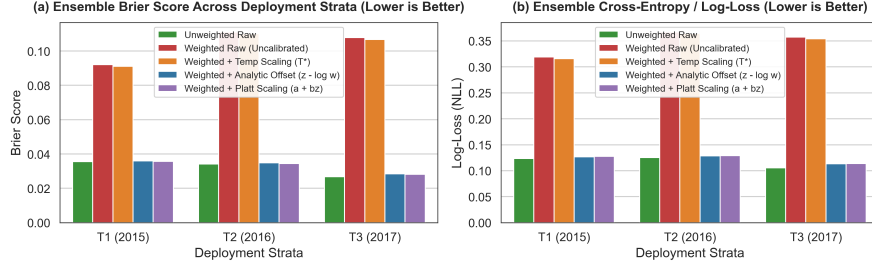


Figure 1: Multi-calibrator evaluation on Pleural Effusion for the DenseNet-121 3-seed ensemble across sequential deidentified-date strata (T_1, T_2, T_3), directly visualizing Table 3. (a) Brier score and (b) Log-loss. Temperature scaling fails to reduce probability error in weighted models, whereas Analytic Offset ($z - \log w$) and Platt Scaling eliminate 74% of Brier error, matching unweighted models.

348 $(FN_{\text{auto}} = \sum_{i \in \text{Auto}} \mathbb{I}(y_i = 1 \wedge \hat{y}_i = 0))$. Note that $TP_{\text{auto}} + FN_{\text{auto}} =$
349 $N_{\text{pos, ret}}$.

- 350 • **Retained Sensitivity:** Diagnostic sensitivity evaluated strictly among
351 the retained automated decisions: $\text{Sens}_{\text{retained}} = \frac{TP_{\text{auto}}}{N_{\text{pos, ret}}} = \frac{TP_{\text{auto}}}{TP_{\text{auto}} + FN_{\text{auto}}}$.
- 352 • **Full-Cohort Triage Sensitivity:** System-wide detection sensitivity
353 across all $N_{\text{pos, total}}$ cases in the evaluation stratum (e.g., $N_{\text{pos, total}} =$
354 85 in T_3), assuming that all examinations deferred to human readers
355 receive expert diagnostic review and are correctly diagnosed: $\text{Sens}_{\text{triage}} =$
356 $\frac{TP_{\text{auto}} + N_{\text{pos, ref}}}{N_{\text{pos, total}}} = 1 - \frac{FN_{\text{auto}}}{N_{\text{pos, total}}}$.
- 357 • **Referral Workload:** The fraction and count of examinations routed
358 to readers ($N_{\text{ref}} = (1 - C) \cdot N$).

359 3. Experimental Results

360 3.1. Experimental Matrix: Discrimination vs. Calibration Trajectories

361 Table 2 presents the comprehensive evaluation of the 2×2 experimental
362 design comprising two architectures and two loss objectives (with three
363 independently trained random seed replications per configuration, 12 models
364 total) for Pleural Effusion, Cardiomegaly, and Pneumonia across chronological
365 strata T_1, T_2, T_3 .

366 *3.1.1. Discrimination Attenuation*

367 Both architectures exhibit moderate attenuation in AUROC across subse-
 368 quent deidentified-date cohort strata. For DenseNet-121 on Pleural Effusion,
 369 AUROC decreases from 0.9335 ± 0.0052 at T_1 to 0.8731 ± 0.0120 at T_3
 370 ($\Delta\text{AUROC} = -0.0604$). ResNet-50 exhibits an identical downward trajectory
 371 ($0.9295 \pm 0.0113 \rightarrow 0.8608 \pm 0.0190$, $\Delta\text{AUROC} = -0.0687$). This confirms
 372 that moderate discriminatory attenuation occurs across subsequent cohort
 373 strata, regardless of the training loss objective.

374 *3.1.2. Temporal Evolution of the Excess Calibration Penalty*

375 In stark contrast to discrimination, probability calibration diverges com-
 376 pletely depending on the training loss function:

- 377 • **Unweighted Models:** Brier scores are exceptionally low and actually
 378 *improve* across subsequent strata for conditions with falling prevalence.
 379 For unweighted DenseNet-121 on Pleural Effusion, the Brier score
 380 improves from 0.0369 ± 0.0029 (T_1) to 0.0279 ± 0.0009 (T_3), while
 381 Equal-Mass ECE decreases from 0.0232 to 0.0090.
- 382 • **Positive-Weighted Models:** Brier scores are dramatically elevated
 383 and appear to degrade across subsequent cohort strata ($0.1081 \pm$
 384 $0.0394 \rightarrow 0.1254 \pm 0.0456$ for DenseNet-121; $0.1256 \pm 0.0325 \rightarrow 0.1392 \pm$
 385 0.0310 for ResNet-50). ECE exceeds 22% across all held-out deidentified-
 386 date strata.

387 To track how this calibration disparity evolves across sequential deidentified-
 388 date cohort strata, we evaluate the excess calibration penalty $\Delta\text{Brier} =$
 389 $\text{Brier}_{\text{weighted}} - \text{Brier}_{\text{unweighted}}$ across sequential strata:

- 390 • **Pleural Effusion (Prevalence Declines: 5.27% $\rightarrow$ 4.44% $\rightarrow$ 3.49%):**
 391 The excess calibration penalty of positive weighting expands mono-
 392 tonically across cohorts, from $\Delta\text{Brier} = +0.0712 \pm 0.0413$ in T_1 to
 393 $+0.0932 \pm 0.0428$ in T_2 and $+0.0975 \pm 0.0454$ in T_3 .
- 394 • **Cardiomegaly (Prevalence Increases: 7.82% $\rightarrow$ 10.58% $\rightarrow$ 10.59%):**
 395 The excess calibration penalty remains stable or slightly attenuates:
 396 $\Delta\text{Brier} = +0.0729 \pm 0.0255$ in T_1 , $+0.0681 \pm 0.0329$ in T_2 , and $+0.0683 \pm$
 397 0.0320 in T_3 .

Table 2: Experimental Matrix Evaluation: Primary Longitudinal Performance across Architectures, Loss Objectives, and Thoracic Pathologies (Strata $T_1 \rightarrow T_3$). Values represent mean $\pm$ standard deviation across three independently trained random seeds ($S \in \{42, 123, 2026\}$). Complete 17-column trajectories across all intermediate deployment strata (T_1, T_2, T_3) and log-loss metrics are detailed in Supplementary Table S3.

Architecture	Objective	Pathology	AUROC ($T_1 \rightarrow T_3$)	Δ AUROC	Brier ($T_1 \rightarrow T_3$)	Δ Brier	ECE T_3
DenseNet-121	Unweighted BCE	Pleural Effusion	0.934 $\rightarrow$ 0.873	-0.060	0.037 $\rightarrow$ 0.028	-0.009	0.0090
DenseNet-121	Unweighted BCE	Cardiomegaly	0.908 $\rightarrow$ 0.902	-0.006	0.053 $\rightarrow$ 0.068	+0.015	0.0210
DenseNet-121	Unweighted BCE	Pneumonia	0.847 $\rightarrow$ 0.865	+0.017	0.024 $\rightarrow$ 0.019	-0.005	0.0095
DenseNet-121	Unweighted BCE	Edema*	0.540 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0029
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	0.906 $\rightarrow$ 0.839	-0.067	0.108 $\rightarrow$ 0.125	+0.017	0.2227
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	0.874 $\rightarrow$ 0.858	-0.016	0.126 $\rightarrow$ 0.136	+0.010	0.1969
DenseNet-121	Pos-Weighted BCE	Pneumonia	0.818 $\rightarrow$ 0.842	+0.024	0.131 $\rightarrow$ 0.130	-0.002	0.2425
DenseNet-121	Pos-Weighted BCE	Edema*	0.612 $\rightarrow$ -	-	0.061 $\rightarrow$ 0.064	+0.002	0.1274
ResNet-50	Unweighted BCE	Pleural Effusion	0.930 $\rightarrow$ 0.861	-0.069	0.037 $\rightarrow$ 0.028	-0.009	0.0132
ResNet-50	Unweighted BCE	Cardiomegaly	0.890 $\rightarrow$ 0.879	-0.011	0.057 $\rightarrow$ 0.071	+0.014	0.0175
ResNet-50	Unweighted BCE	Pneumonia	0.843 $\rightarrow$ 0.880	+0.037	0.024 $\rightarrow$ 0.019	-0.005	0.0106
ResNet-50	Unweighted BCE	Edema*	0.556 $\rightarrow$ -	-	0.002 $\rightarrow$ 0.000	-0.002	0.0017
ResNet-50	Pos-Weighted BCE	Pleural Effusion	0.879 $\rightarrow$ 0.819	-0.060	0.126 $\rightarrow$ 0.139	+0.014	0.2751
ResNet-50	Pos-Weighted BCE	Cardiomegaly	0.830 $\rightarrow$ 0.799	-0.031	0.142 $\rightarrow$ 0.152	+0.011	0.2338
ResNet-50	Pos-Weighted BCE	Pneumonia	0.835 $\rightarrow$ 0.823	-0.012	0.167 $\rightarrow$ 0.169	+0.002	0.3253
ResNet-50	Pos-Weighted BCE	Edema*	0.549 $\rightarrow$ -	-	0.069 $\rightarrow$ 0.075	+0.005	0.1670

• **Pneumonia (Prevalence Steady Low: 2.66% $\rightarrow$ 1.98% $\rightarrow$ 2.05%):**

The penalty remains consistently elevated: Δ Brier = $+0.1079 \pm 0.0526$ in T_1 , $+0.1126 \pm 0.0552$ in T_2 , and $+0.1112 \pm 0.0508$ in T_3 .

This establishes the operational mechanism: a static training-induced positive logit offset produces surplus false-positive probability mass that is penalized more severely when deployment disease prevalence drops, expanding the apparent temporal calibration gap.

3.2. Multi-Calibrator Comparison

Table 3 and Fig. 1 present the multi-calibrator benchmark for Pleural Effusion on the 3-seed ensemble across 2,000 patient-clustered bootstrap resamples, including Cox calibration regression intercept (α) and slope (β) parameters.

3.2.1. Temperature Scaling Analysis

For weighted DenseNet-121, temperature scaling optimizes $T^* \approx 1.0$, producing an uncalibrated T_3 Brier score of 0.1067 compared to raw 0.1079. The paired bootstrap contrast on the ensemble is Δ Brier = -0.0011 [95% CI: $-0.0014, -0.0009$], with the calibration intercept remaining severely displaced ($\alpha = -2.9615$ vs. raw -2.9638). For ResNet-50, temperature scaling achieves a modest variance adjustment (Δ Brier = -0.0220 [95% CI:

Table 3: Multi-Calibrator Comparison on Pleural Effusion: Deidentified-Date Ensemble Performance, Calibration Intercept (α), Slope (β), and Paired Bootstrap Contrasts. Complete 12-column factorial evaluation across all intermediate strata and log-loss is provided in Supplementary Table S4.

Architecture	Loss Objective	Calibrator	Brier ($T_1 \rightarrow T_3$)	T_3 α	T_3 β	T_3 ECE	T_3 Δ Brier vs Raw [95% CI]
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.092 $\rightarrow$ 0.108	-2.964	1.048	0.2227	—
DenseNet-121	Weighted BCE	Temperature Scaling (T^*)	0.091 $\rightarrow$ 0.107	-2.962	1.029	0.2190	-0.0011 [-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.036 $\rightarrow$ 0.028	-0.445	0.929	0.0126	-0.0795 [-0.0893, -0.0697]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.036 $\rightarrow$ 0.028	-0.229	1.128	0.0176	-0.0798 [-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	+0.147	1.044	0.0040	—
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.110 $\rightarrow$ 0.123	-2.853	1.950	0.2751	—
ResNet-50	Weighted BCE	Temperature Scaling (T^*)	0.088 $\rightarrow$ 0.101	-2.608	1.184	0.1993	-0.0220 [-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.040 $\rightarrow$ 0.029	+1.075	1.405	0.0158	-0.0936 [-0.1022, -0.0844]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.038 $\rightarrow$ 0.030	-0.191	1.178	0.0223	-0.0930 [-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw Baseline	0.036 $\rightarrow$ 0.027	-0.077	1.095	0.0109	—

417 $-0.0241, -0.0198]$), but leaves the underlying intercept discrepancy uncor-
 418 rected ($\alpha = -2.6080$ vs. raw -2.8526). This demonstrates that temperature
 419 scaling provides limited or architecture-dependent empirical scaling, but can-
 420 not resolve the structural intercept displacement induced by positive loss
 421 weighting.

422 3.2.2. Offset Correction and Parity Restoration

423 The training positive weight for Pleural Effusion is $w = 21.51$, yielding
 424 an analytic logit offset of $\log w = +3.068$. Applying the zero-parameter
 425 subtraction $z - \log w$ directly drops the T_3 Brier score from 0.1079 to 0.0284
 426 (Δ Brier = -0.0795 [95% CI: $-0.0893, -0.0697$], with the 2,000-replicate
 427 patient-clustered bootstrap interval strictly excluding zero), reducing ECE
 428 from 22.3% to 1.26%. Analytic correction substantially reduced intercept
 429 miscalibration and brought the Brier score close to the unweighted baseline,
 430 although residual intercept miscalibration remained architecture-dependent:
 431 for DenseNet-121, the Cox calibration intercept moves from $\alpha = -2.9638$
 432 directly to $\alpha = -0.4451$ while perfectly preserving the calibration slope
 433 ($\beta = 0.9292$), whereas for ResNet-50 the corrected intercept is $\alpha = +1.0751$.

434 Fitted Offset ($z + a^*$) and Platt Scaling achieve nearly identical performance
 435 (T_3 Brier: 0.0289 and 0.0281; Δ Brier = -0.0790 [95% CI: $-0.0885, -0.0696$]
 436 and -0.0798 [95% CI: $-0.0894, -0.0702]$). The zero-parameter analytic cor-
 437 rection closely approaches a flexible two-parameter Platt model without fitting
 438 a slope or intercept. For ResNet-50, Analytic Offset similarly eliminates 76%
 439 of probability error (T_3 Brier: $0.1227 \rightarrow 0.0291$, Δ Brier = -0.0936 [95% CI:
 440 $-0.1022, -0.0844]$), confirming that intercept neutralization is architecturally
 441 universal.

442 3.2.3. Empirical Verification of Logit Offset: $z_w - z_u \approx \log w$

443 To directly validate the theoretical prediction of Equation (8) in finite
 444 deep neural networks, Fig. 2 plots the empirical distributions of logit differ-
 445 ences $\Delta z(x) = \bar{z}_w(x) - \bar{z}_u(x)$ evaluated across all 16,604 image-architecture
 446 ensemble pairs (averaged across the three seed-matched model pairs) against
 447 the theoretical reference value $\log w$:

- 448 • **Pleural Effusion** ($\log w = 3.07$): The observed distribution clusters
 449 tightly around the theoretical value, with empirical mean $\Delta z = 3.29 \pm$
 450 0.81 and median 3.25 (interquartile range [IQR]: 2.77 – 3.79), exhibiting
 451 an average bias of only $+0.22$ logit units.
- 452 • **Pneumonia** ($\log w = 3.99$): Observed mean $\Delta z = 3.83 \pm 0.74$ and
 453 median 3.84 (IQR: 3.34 – 4.32), closely matching theory with a minimal
 454 discrepancy of -0.16 logit units.
- 455 • **Cardiomegaly** ($\log w = 2.25$): Observed mean $\Delta z = 2.64 \pm 1.04$ and
 456 median 2.60 (IQR: 1.91 – 3.31), bias $+0.39$ logit units.
- 457 • **Edema (Extreme Imbalance: $w = 1091.3, \log w = 7.00$)**: With only
 458 23 positive cases in the training set, empirical logit shifts saturate at
 459 mean $\Delta z = 3.69 \pm 1.47$ (median 3.46 , IQR: 2.63 – 4.58). Here, L_2 weight
 460 regularization and finite gradient updates prevent network outputs from
 461 attaining the theoretical $\approx +7.0$ asymptote, highlighting the finite-
 462 sample boundary of the asymptotic derivation.

463 3.2.4. Reliability Diagrams

464 Figure 3 shows reliability diagrams across sequential deidentified-date
 465 strata T_1, T_2, T_3 using adaptive 10-bin equal-mass partitioning. Raw weighted
 466 models display severe over-confidence across all probability deciles. Applying
 467 the analytic offset restores the curves tightly onto the ideal diagonal across
 468 all deployment cohorts, identical to unweighted models.

469 3.3. Selective Prediction and Workload Triage Simulation

470 Table 4 and Fig. 4 summarize the selective triage simulation on the later
 471 held-out deidentified-date stratum T_3 ($N = 2,436$ radiographs, containing
 472 85 true positive Pleural Effusion cases), explicitly accounting for retained
 473 positives, automated true positives, missed cases, and both retained and
 474 full-cohort triage sensitivities.

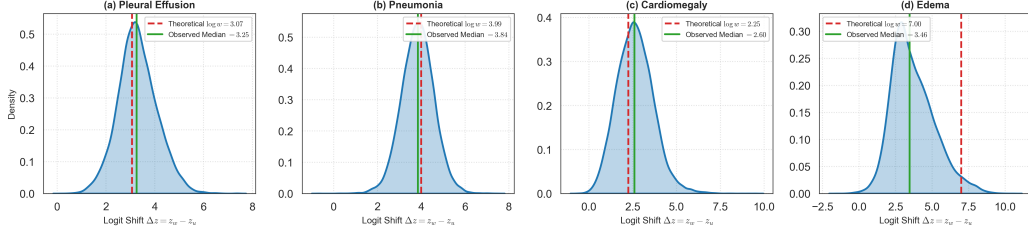


Figure 2: Empirical verification of theoretical logit shift ($z_w - z_u \approx \log w$) across 16,604 held-out deidentified-date image evaluations (8,302 images $\times$ 2 architectures, with Δz computed as the mean difference across the three seed-matched weighted/unweighted model pairs). Kernel density estimates (blue) of observed logit differences between positive-weighted and unweighted models cluster tightly around the theoretical population attractor $\log w$ (dashed red line; observed median in solid green) across Pleural Effusion (a), Pneumonia (b), and Cardiomegaly (c). For Edema (d), extreme training class scarcity ($w > 1000$) induces optimization saturation, bounding empirical shifts near $+3.69$.

At 100% coverage (full automation), the raw weighted model incurs 16.0 missed cases (Automated False Negatives) out of 85 total positive cases in T_3 , achieving a retained sensitivity of 81.18%, AUROC of 0.8691, and Brier score of 0.1079. Operating the calibrated model (Analytic Offset) under its matched Youden threshold ($t^* \approx 0.022$) yields identical sensitivity (81.18%) and 16.0 missed cases, but with an almost four-fold lower Brier score of 0.0284. The unweighted raw model achieves 12.0 missed cases (sensitivity 85.88%, AUROC 0.8806, Brier 0.0268).

Under selective triage based on prediction margin from each method’s operational decision cutoff:

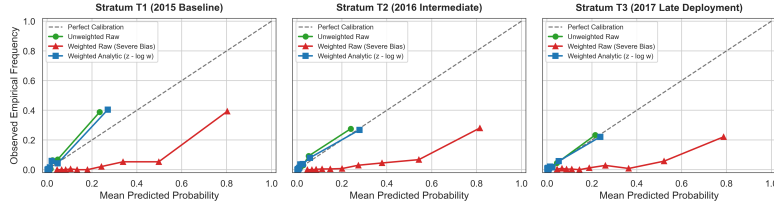


Figure 3: Reliability diagrams for Pleural Effusion (DenseNet-121 3-seed ensemble) across held-out deidentified-date strata (T_1, T_2, T_3). Raw weighted models (red) severely over-predict event probabilities across all bins. Analytic offset calibration (blue) and unweighted models (green) remain tightly calibrated to the ideal diagonal across all deployment windows.

Table 4: Selective Prediction and Reader Referral Triage Simulation on Later Held-Out Stratum T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases). Evaluated under matched Youden decision thresholds determined on development stratum T_1 across primary clinical operating regimes (100%, 90%, 80%, 70% coverage). Complete 24-row simulation across all coverage levels is detailed in Supplementary Table S5.

Coverage	Referred (Workload)	Strategy	Automated FN (Missed)	Retained Sens.	Full Sens.	Retained AUROC	Retained Brier
100%	0 (0%)	Weighted BCE (Raw)	16.0	81.18%	81.18%	0.8691	0.1079
100%	0 (0%)	Weighted BCE (Analytic Corrected)	16.0	81.18%	81.18%	0.8623	0.0284
100%	0 (0%)	Unweighted BCE (Raw)	12.0	85.88%	85.88%	0.8806	0.0268
90%	243 (10%)	Weighted BCE (Raw)	14.0	82.93%	83.53%	0.8789	0.1062
90%	243 (10%)	Weighted BCE (Analytic Corrected)	15.0	81.01%	82.35%	0.8735	0.0289
90%	243 (10%)	Unweighted BCE (Raw)	12.0	84.62%	85.88%	0.8923	0.0266
80%	487 (20%)	Weighted BCE (Raw)	10.0	87.01%	88.24%	0.8882	0.1012
80%	487 (20%)	Weighted BCE (Analytic Corrected)	15.0	77.61%	82.35%	0.8896	0.0264
80%	487 (20%)	Unweighted BCE (Raw)	12.0	82.61%	85.88%	0.9027	0.0256
70%	730 (30%)	Weighted BCE (Raw)	7.0	89.39%	91.76%	0.8997	0.0956
70%	730 (30%)	Weighted BCE (Analytic Corrected)	13.0	70.45%	84.71%	0.8876	0.0178
70%	730 (30%)	Unweighted BCE (Raw)	11.0	78.00%	87.06%	0.9051	0.0194
70%	730 (30%)	Random Deferral Baseline	11.3	80.99%	86.69%	—	0.1081

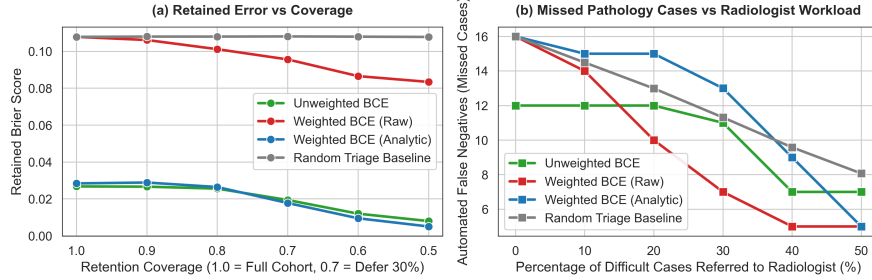


Figure 4: Selective prediction evaluation on later held-out stratum T_3 with matched development thresholds (Table 4). Left: Risk-coverage curves demonstrating that selective deferral steadily reduces Brier error across all models as ambiguous cases are referred. Right: Reader referral workload vs. automated false negatives, illustrating the distinct operational behaviors of raw weighted models (halving missed cases to 7.0 under 30% referral) and analytic corrected models (optimizing probability accuracy with Brier 0.0178).

- 485 • **Raw Weighted BCE (Triage-Optimized):** At 90% coverage (243

486 borderline radiographs referred to a reader), automated false negatives

487 decrease to 14.0 (full-cohort system sensitivity 83.53%). At 80% cov-

488 erage (487 referred), false negatives drop to 10.0 (system sensitivity

489 88.24%). At 70% coverage (730 examinations referred, 30% workload),

490 66 true positive cases remain in the automated fraction and 59 are

491 correctly flagged as positive, cutting missed cases by more than half to

492 7.0 (retained sensitivity 89.39%, full-cohort triage sensitivity 91.76%),

493 outperforming random triage (11.3 missed cases). However, retained

494 probabilities remain uncalibrated and biased high (Brier score 0.0956).

- 495 • **Analytic Corrected BCE (Probability-Optimized):** Operating
496 under its matched threshold ($t^* \approx 0.022$), the calibrated model achieves
497 an exceptional Brier score of 0.0178 at 70% coverage (a > 5 -fold reduc-
498 tion in probability error relative to the raw model). However, because
499 predicted probabilities are concentrated near true clinical prevalence
500 ($p \in [0.001, 0.05]$), the distance normalization in Equation (12) assigns
501 low uncertainty to low-probability false negatives ($\hat{p} \ll t^*$). Conse-
502 quently, subtle false negatives are retained rather than referred, leaving
503 13.0 automated false negatives among 44 retained positives (retained
504 sensitivity 70.45%, full-cohort triage sensitivity 84.71%) at 70% cover-
505 age.
- 506 • **Raw Unweighted BCE:** At 70% coverage, the unweighted model
507 retains 50 positive cases, correctly flags 39, and incurs 11.0 automated
508 false negatives with retained Brier score 0.0194 (retained sensitivity
509 78.00%, full-cohort triage sensitivity 87.06%).

510 Importantly, these empirical results expose an operational trade-off in ma-
511 chine learning deployment: post-hoc probability calibration (Analytic Offset)
512 is essential for accurate risk estimation and prognosis, dropping Brier scores to
513 near-optimal levels ($0.0284 \rightarrow 0.0178$); conversely, threshold-margin selective
514 triage on raw positive-weighted models is effective for recall-oriented screening
515 referral (cutting missed cases from 16 to 7), but yields biased probabilities.
516 Because the analytic logit transformation $z \mapsto z - \log w$ is strictly monotonic,
517 any ranking-based decision rule under an equivalent quantile threshold pro-
518 duces identical patient prioritization. The operational divergence observed
519 in Table 4 arises specifically because the uncertainty triage metric $u(x)$ in
520 Equation (12) measures distance relative to the operational decision cutoff t^*
521 determined on development stratum T_1 . For calibrated models, $t^* \approx 0.022$
522 aligns with low clinical prevalence, causing low-probability false negatives
523 ($\hat{p} \ll t^*$) to exhibit low relative uncertainty and remain in the automated
524 pool; for raw weighted models, the elevated threshold ($t^* \approx 0.33$) places
525 borderline cases within the referral margin. Consequently, selective referral
526 performance is fundamentally policy-dependent: calibrated probabilities are
527 required for valid decision-theoretic risk stratification, whereas raw weighted
528 models with threshold-margin triage act as a heuristic for recall-oriented
529 screening. Crucially, neither system achieves zero false negatives, refuting
530 unsubstantiated claims of complete error elimination in autonomous settings.

3.4. Cross-Institutional External Evaluation (MIMIC-CXR-JPG)

The external MIMIC-CXR-JPG cohort comprised 3,403 frontal radiographs from 3,041 studies and 289 patients, with complete image availability. Rather than asserting turnkey diagnostic transportability—as confirmed by moderate discrimination for Pleural Effusion (AUROC 0.688 [95% CI: 0.656, 0.717] for unweighted DenseNet-121 and 0.677 [95% CI: 0.645, 0.709] for unweighted ResNet-50), weak discrimination for Pneumonia (0.583–0.607) and Edema (0.516–0.581), and near-chance discrimination for Cardiomegaly (0.519–0.525, with patient-clustered 95% CIs spanning 0.50)—this experiment functions as a rigorous external stress-test of whether loss-induced logit distortion and zero-parameter offset recovery persist across distinct health systems, acquisition hardware, and report extraction pipelines. Negative Brier skill scores ($BSS < 0$) across all configurations indicated that transported BRAX probabilities were less accurate than a local MIMIC prevalence-only reference model ($Brier_{\text{null}} = \pi(1 - \pi)$), demonstrating substantial cross-institutional baseline calibration shift.

Evaluating paired analytic offset recalibration ($\bar{p}_{\text{corrected}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on positive-weighted models demonstrated statistically significant reductions in Brier error for Cardiomegaly ($\Delta Brier = -0.0579$ [95% CI: $-0.0790, -0.0349$] on DenseNet-121; -0.0611 [95% CI: $-0.0818, -0.0415$] on ResNet-50) and Pneumonia ($\Delta Brier = -0.0095$ [95% CI: $-0.0170, -0.0016$] on DenseNet-121; -0.0155 [95% CI: $-0.0254, -0.0059$] on ResNet-50), with consistent point-estimate error reduction on Pleural Effusion (-0.0084 and -0.0109). On MIMIC-CXR-JPG, subtraction of the full theoretical offset worsened the Edema Brier score. This result is consistent with over-correction when the empirically realized finite-network shift is smaller than the theoretical $\log w$; however, cross-institutional covariate and label differences prevent attribution of this external result uniquely to optimization saturation. Because logit subtraction $z - \log w$ is strictly monotonic, discriminatory ranking (AUROC) was perfectly preserved across all pathologies.

To evaluate sensitivity to repeated within-patient examinations and multi-image studies, we conducted a parallel sensitivity analysis aggregated strictly at the diagnostic study level ($N = 3,041$ studies across $N = 289$ patients) using maximum probability pooling ($\max_i p_i$). Study-level discrimination was concordant with image-level evaluation across all pathologies: Pleural Effusion AUROC was 0.695 [95% CI: 0.663, 0.723] for unweighted DenseNet-121 and 0.682 [95% CI: 0.650, 0.714] for ResNet-50; Cardiomegaly AUROC was 0.526 [95% CI: 0.497, 0.559] and 0.521 [95% CI: 0.494, 0.548]; Pneumonia AUROC

Table 5: Cross-Institutional External Validation on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Images from 3,041 Studies Across $N = 289$ Unique Patients). Primary experimental evaluation of paired zero-parameter analytic logit recalibration ($\bar{p}_{\text{raw}} = \frac{1}{3} \sum \sigma(z_s) \rightarrow \bar{p}_{\text{corr}} = \frac{1}{3} \sum \sigma(z_s - \log w)$) on 3-seed ensembles trained with positive-weighted BCE. Evaluated with empirical prevalence (π), local null reference baseline ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), patient-clustered 1,000-replicate bootstrap 95% confidence intervals, Brier skill score ($\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{null}}$), and paired contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$). The complete 16-row factorial comparison with unweighted BCE baselines is detailed in Supplementary Table S1, and study-level sensitivity analysis is detailed in Supplementary Table S2.

Architecture	Pathology	Prevalence (π)	Brier _{null}	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

was 0.608 [95% CI: 0.562, 0.650] and 0.584 [95% CI: 0.543, 0.624]; and Edema AUROC was 0.565 [95% CI: 0.528, 0.600] and 0.516 [95% CI: 0.480, 0.553], with study-level null Brier baselines of 0.2196 (Effusion), 0.1951 (Cardiomegaly), 0.0913 (Pneumonia), and 0.1697 (Edema). Furthermore, paired analytic recalibration on positive-weighted models demonstrated identical calibration invariance at study level: statistically significant Brier reductions were reproduced for Cardiomegaly ($\Delta\text{Brier} = -0.0601$ [95% CI: $-0.0814, -0.0367$] on DenseNet-121; -0.0642 [95% CI: $-0.0852, -0.0434$] on ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0100$ [$-0.0179, -0.0018$]; -0.0163 [$-0.0267, -0.0068$]), with consistent point-estimate reductions for Pleural Effusion (-0.0102 and -0.0118 ; Supplementary Table S2). This qualitative concordance confirms that cross-institutional discrimination patterns and recalibration effects are preserved when aggregating by diagnostic study.

4. Discussion

4.1. Disentangling Loss-Induced Probability Distortion from Deidentified-Date Cohort Shift

A substantial body of recent literature in biomedical informatics has sounded alarms regarding temporal calibration decay in healthcare AI [8, 9]. Our findings elucidate the mathematical and empirical mechanisms underlying these observations. It is critical to distinguish between discrimination

loss and calibration error: our experiments confirm that moderate discrimination attenuation does occur across later deidentified-date cohort strata (AUROC decreases by approximately 0.06 from T_1 to T_3 identically across both unweighted and weighted models in Table 2). However, the simultaneous perception of severe, progressive calibration collapse is largely an artifact of positive loss weighting compounded by shifting disease prevalence.

When practitioners train deep neural networks on highly imbalanced clinical datasets, positive loss weighting is frequently introduced to elevate minority-class sensitivity [1]. However, as derived in Equation (8) and verified empirically in Fig. 2, this introduces a static logit shift $\approx \log w$. When evaluated on later cohort strata where disease prevalence shifts (e.g., Pleural Effusion prevalence declining from 5.27% in T_1 to 3.49% in T_3), this fixed positive offset generates surplus false-positive probability mass that compounds the apparent Brier error. Because standard post-hoc evaluation pipelines rely almost exclusively on temperature scaling—which cannot alter the logit intercept—evaluators observing deteriorating Brier scores have routinely attributed the failure to intrinsic model decay.

Our experiments provide direct evidence against this attribution. When the mathematical offset is neutralized ($z - \log w$), the apparent calibration decay vanishes: Brier scores decrease across subsequent strata, moving in lockstep with the unweighted baseline as disease prevalence drops. Thus, while discrimination attenuation reflects underlying dataset shift across cohort strata, severe calibration degradation is predominantly an uncorrected post-training calibration bias that can be decoupled and resolved analytically.

4.2. Relationship to Prior Work on Class Weights and Likelihood Recovery

Our theoretical analysis connects directly with the foundational work of Caplin, Martin, and Marx [14], who demonstrated that training with class weights inherently corrupts probability calibration and established an inverse modeling method to recover undistorted likelihoods, illustrating their recovery technique on a static CheXNet pneumonia model. While Caplin et al. established that class weighting and raw calibration are fundamentally at odds in stationary settings, our study addresses critical dimensions left unexplored in prior theoretical treatments:

1. **Resolution of the “Temporal Calibration Decay” Paradox:** Caplin et al. investigated stationary, single-split validation. We address the active clinical informatics debate regarding longitudinal stability [8, 9, 12], proving that apparent temporal calibration decay over

chronological cohort strata ($T_1 \rightarrow T_3$) is driven by uncorrected training-loss offsets interacting with dynamic clinical prevalence rather than intrinsic temporal drift.

2. **Structural Failure of Temperature Scaling:** While Caplin et al. focused on recovering likelihoods from weighted models, standard clinical vision pipelines continue to rely on temperature scaling [15]. We provide analytical and empirical proof that temperature scaling is structurally incapable of rectifying weighted models because it operates solely on logit dispersion while leaving the large negative calibration intercept ($\alpha \approx -2.96$) completely uncorrected.
3. **Finite Neural Network Dynamics and Scarcity Bounds:** Unlike idealized asymptotic formulations, our 16,604 paired evaluations across DenseNet-121 and ResNet-50 (spanning 49,812 seed-level model runs) uncover where finite deep networks diverge from theory: while empirical shifts match $\log w$ across moderate-prevalence conditions, extreme class scarcity ($w > 1000$ in Edema) induces optimization saturation ($\Delta z \approx +3.69$ vs. theoretical $+7.00$) due to L_2 regularization and gradient clipping.
4. **Cross-Institutional Stress-Testing:** Rather than asserting turnkey clinical transportability (given modest to near-chance external discrimination), we evaluate whether zero-parameter analytic logit correction ($z - \log w$) survives extreme multi-center stress-testing from BRAX ($N = 40,523$, Brazil) to MIMIC-CXR ($N = 3,403$, USA), delineating where loss correction successfully mitigates calibration error versus where cross-institutional domain shifts dominate.
5. **Clinical Decision Triage vs. Probability Calibration:** Extending recent multi-dataset selective CXR investigations [22], we operationalize the practical dilemma facing clinicians: while analytic correction ($z - \log w$) is strictly required for epidemiological risk stratification and prognosis (restoring Brier parity, consistent with plug-in probability recovery principles [18]), raw weighted models provide superior threshold-margin selective triage by halving automated false negatives under operational referral constraints.

4.3. Architectural Generalizability

The observed phenomena are remarkably consistent across distinct deep learning architectures. Both DenseNet-121 (dense convolutional concate-

662 nation) and ResNet-50 (residual identity mapping) exhibit near-identical
 663 behavior:

- 664 • Discrimination attenuation across later held-out strata is virtually indis-
 665 tinguishable between architectures ($\Delta\text{AUROC} = -0.0604$ vs. -0.0687).
- 666 • Both architectures experience severe raw calibration error under weighted
 667 BCE (raw ensemble Brier 0.1079 vs 0.1227 on T_3 ; 0.1254 ± 0.0456 vs
 668 0.1392 ± 0.0310 across individual seeds), characterized by large negative
 669 calibration intercepts ($\alpha \approx -2.96$ and -2.85).
- 670 • For both architectures, Analytic Offset reduces total Brier score by ap-
 671 proximately 74% to 76% ($\Delta\text{Brier} = -0.0795$ [95% CI: $-0.0893, -0.0697$]
 672 for DenseNet-121 and -0.0936 [95% CI: $-0.1022, -0.0844$] for ResNet-
 673 50), substantially reducing intercept miscalibration ($\alpha = -0.4451$
 674 and $+1.0751$) while preserving calibration slopes ($\beta = 0.9292$ and
 675 1.4052). This confirms that logit-shift mechanics are universal across
 676 convolutional vision backbones, though residual intercept bias remains
 677 architecture-dependent.

678 4.4. Simulated Human-in-the-Loop Triage and Operational Trade-offs

679 In real-world clinical deployment workflows, autonomous decision-making
 680 must be evaluated through uncertainty-aware deferral simulations:

- 681 1. **Risk Stratification and Prognosis:** Well-calibrated models are es-
 682 sential for automated rule-out and downstream risk scoring. Applying
 683 Analytic Offset ($z - \log w$) drops Brier error to 0.0178 at 70% cover-
 684 age, restoring reliable posterior probabilities aligned with true clinical
 685 prevalence.
- 686 2. **High-Sensitivity Screening Referral:** In high-throughput screening
 687 where the primary objective is minimizing missed disease, threshold-
 688 margin selective triage on raw positive-weighted models cuts automated
 689 false negatives from 16 to 7 (full-cohort triage sensitivity 91.76%) under
 690 a 30% reader referral budget. However, because probabilities remain
 691 biased high (Brier 0.0956), these outputs should not be used as calibrated
 692 risks.
- 693 3. **Zero-Parameter Operational Utility:** The analytic offset ($z - \log w$)
 694 requires zero retraining, zero validation optimization, and zero compute
 695 overhead, enabling immediate integration on edge inference devices.

696 4.5. *Scope and Boundary of Findings and Limitations*

697 To establish the scope and boundary of findings, several important method-
698 ological and clinical limitations should be highlighted:

- 699 1. **Report-Derived Label Noise:** Ground-truth labels in BRAX were
700 generated via natural language processing from Portuguese text reports.
701 Although validated against expert consensus, report-derived labels carry
702 inherent extraction noise compared to dedicated multi-radiologist image
703 re-reading.
- 704 2. **Cross-Institutional Domain Gap and Modest Diagnostic Trans-**
705 **portability:** Our external evaluation on MIMIC-CXR-JPG serves to
706 stress-test calibration mechanics under severe domain shift rather than
707 demonstrate ready-to-deploy clinical diagnostic transportability. Mod-
708 els achieved only modest discrimination for Pleural Effusion (AUROC
709 ≈ 0.69) and degraded near chance for Cardiomegaly (0.52) and Pneu-
710 monia (0.59), underscoring that while mathematical offset correction
711 improves calibration for several targets, real-world multi-site clinical
712 deployment demands local fine-tuning or domain-invariant feature rep-
713 resentations.
- 714 3. **Date Anonymization and Temporal Semantics:** BRAX docu-
715 mentation states that the released dates are fictitious dates generated
716 through a hashing procedure while retaining the original intervals be-
717 tween study acquisitions. Because synchronization of anonymized dates
718 across different patients is not explicitly established, we interpret these
719 partitions as sequential deidentified-date strata rather than real-world
720 calendar periods. While within-patient chronological ordering is pre-
721 served and patient-clustered bootstrapping ensures valid statistical
722 inference, these sequential strata (T_1, T_2, T_3) capture longitudinal shift
723 across deidentified strata rather than synchronized calendar years.
- 724 4. **Retrospective Simulation vs. Real-Time Deployment:** This
725 study constitutes a retrospective simulation on frozen historical cohorts.
726 Future real-time clinical deployment requires silent shadow evaluation
727 within operational hospital Picture Archiving and Communication Sys-
728 tems (PACS).
- 729 5. **Architectural Scope:** We evaluated two widely established convolu-
730 tional neural network backbones (DenseNet-121 and ResNet-50). Al-
731 though our mathematical derivations apply to binary cross-entropy
732 minimization generally, evaluating modern Vision Transformers (ViT)

and multimodal vision-language models represents an important direction for future research.

6. **Optimization Randomness and Seed Count:** Experiments were conducted across three independently trained random seeds ($S \in \{42, 123, 2026\}$). While inter-seed standard deviations were reported alongside patient-clustered bootstrap confidence intervals, larger ensembles across diverse initialization trajectories would further refine empirical confidence bounds.

7. **Simulated Reader Triage without Interactive Human Study:** Our selective prediction simulation assumes that examinations deferred to human readers are diagnosed with 100% sensitivity. In real clinical environments, human radiologist sensitivity varies with reader fatigue, experience level, and interpretive setting. An actual clinician-in-the-loop reader study is needed to evaluate true joint human-AI diagnostic accuracy.

8. **Extreme Class Scarcity Pathologies:** For conditions with extreme class scarcity in training (e.g., Edema with only 23 positive training cases, $w = 1091.3$), finite optimization dynamics, L_2 weight decay, and gradient clipping saturate empirical logit shifts at $\Delta z \approx +3.69$ rather than the theoretical $+7.00$ asymptote. Specialized extreme-imbalance formulations are required in such scarcity regimes.

5. Conclusion

In this controlled BRAX experiment, most severe calibration error was associated with positive loss weighting rather than measurable deterioration across later deidentified-date strata. Because standard temperature scaling cannot adjust the logit intercept, it leaves this structural distortion intact. A zero-parameter analytic offset ($z - \log w$) or non-negative Platt scaling reduces total Brier score by approximately 74% to 76% (removing approximately 98% of the Brier-score gap relative to the unweighted baseline), substantially reducing intercept miscalibration and restoring calibration parity with unweighted models without model retraining or validation data. Furthermore, selective prediction simulations with matched decision thresholds reveal an operational dichotomy: raw positive-weighted models provide effective recall-oriented triage by halving automated missed cases under a 30% reader referral budget, whereas analytic calibration restores epidemiologically faithful probabilities for downstream clinical risk stratification. Neither system eliminates missed

disease entirely, underscoring the necessity of transparent, workload-aware deployment policies in computational medicine.

CRedit Authorship Contribution Statement

Vishesh Panghal: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project Administration.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The BRAX dataset (v1.1.0) [2] is available on PhysioNet at <https://physionet.org/content/brax/1.1.0/> under the PhysioNet Credentialed Data Use Agreement. The MIMIC-CXR-JPG database (v2.1.0) [23, 24] is available on PhysioNet at <https://physionet.org/content/mimic-cxr-jpg/2.1.0/> under the PhysioNet Credentialed Data Use Agreement. Access requires credentialed PhysioNet researcher status, completion of human subjects research training, and signed data use agreements. Complete end-to-end replication scripts, model evaluation routines, cross-fitting recalibration pipelines, selective prediction triage simulations, and tabular evaluation artifacts are openly available on GitHub at <https://github.com/Vishesh-panghal/temporal-brax-calibration>, and the repository will be permanently assigned a persistent DOI via Zenodo upon acceptance of the manuscript. No credentialed patient-level images, metadata, labels, identifiers, or prediction rows are redistributed; the repository contains aggregate evaluation artifacts and a synthetic manifest schema example.

Ethical Approval and Consent to Participate

This study represents a secondary analysis of credentialed-access, de-identified diagnostic imaging databases accessible under formal Data Use Agreements. The primary BRAX dataset collection was approved by the

799 Institutional Review Board (IRB) of Hospital Israelita Albert Einstein (São
800 Paulo, Brazil; CAAE: 36720520.1.0000.0071) with a formal waiver of informed
801 consent. Creation and distribution of the MIMIC-CXR-JPG database was
802 approved by the Institutional Review Boards of Beth Israel Deaconess Medical
803 Center (Boston, MA) and the Massachusetts Institute of Technology (Protocol
804 #0403000206) with a waiver of informed consent. The author completed
805 mandatory human subjects research training (CITI Program certification),
806 executed approved PhysioNet Data Use Agreements for both databases, and
807 strictly complied with data governance protocols prohibiting re-identification
808 or redistribution of protected health information.

809 Declaration of Generative AI and AI-Assisted Technologies in the 810 Writing Process

811 During the preparation of this work, the author utilized generative artificial
812 intelligence platforms (Google Gemini, Anthropic Claude, and OpenAI
813 Codex) exclusively for LaTeX formatting and syntax verification, analysis code
814 refactoring and modularization, grammatical refinement, and bibliographic
815 cross-referencing. Following the use of these tools, the author thoroughly
816 reviewed, verified, and edited all text, numerical outputs, and code, and takes
817 full intellectual and scientific responsibility for the integrity, validity, and
818 conclusions of the published manuscript.

819 Funding

820 This research did not receive any specific grant from funding agencies in
821 the public, commercial, or not-for-profit sectors.

822 References

- 823 [1] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding,
824 A. Bagul, C. Langlotz, K. Shpanskaya, et al., CheXNet: Radiologist-level
825 pneumonia detection on chest X-rays with deep learning, arXiv preprint
826 arXiv:1711.05225 (2017).
- 827 [2] E. P. Reis, J. P. Q. de Paiva, M. C. B. da Silva, G. A. S. Ribeiro,
828 V. F. Paiva, L. Bulgarelli, H. M. H. Lee, P. V. Santos, V. M. Brito,
829 L. T. W. Amaral, G. L. Beraldo, J. N. Haidar Filho, G. B. S. Teles,
830 G. Szarf, T. Pollard, A. E. W. Johnson, L. A. Celi, E. Amaro, BRAX, a

- 831 Brazilian labeled chest X-ray dataset, *Scientific Data* 9 (1) (2022) 487.
832 [doi:10.1038/s41597-022-01608-8](https://doi.org/10.1038/s41597-022-01608-8).
- 833 [3] S. G. Finlayson, A. Subbaswamy, K. Singh, J. Bowers, A. Kupke, J. Zit-
834 train, I. S. Kohane, S. Saria, The clinician and dataset shift in artificial
835 intelligence, *New England Journal of Medicine* 385 (3) (2021) 283–286.
836 [doi:10.1056/NEJMc2104626](https://doi.org/10.1056/NEJMc2104626).
- 837 [4] A. Kore, E. A. Babil, V. Subasri, M. Abdalla, B. Fine, E. Dolatabadi,
838 M. Abdalla, Empirical data drift detection experiments on real-world
839 medical imaging data, *Nature Communications* 15 (1) (2024) 1887. [doi:](https://doi.org/10.1038/s41467-024-46142-w)
840 [10.1038/s41467-024-46142-w](https://doi.org/10.1038/s41467-024-46142-w).
- 841 [5] M. Shenouda, I. Flerlage, A. Kaveti, M. L. Giger, S. G. Armato, III,
842 Assessment of a deep learning model for COVID-19 classification on
843 chest radiographs: a comparison across image acquisition techniques
844 and clinical factors, *Journal of Medical Imaging* 10 (6) (2023) 064504.
845 [doi:10.1117/1.jmi.10.6.064504](https://doi.org/10.1117/1.jmi.10.6.064504).
- 846 [6] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, E. W.
847 Steyerberg, Calibration: the Achilles heel of predictive analytics, *BMC*
848 *Medicine* 17 (1) (2019) 230. [doi:10.1186/s12916-019-1466-7](https://doi.org/10.1186/s12916-019-1466-7).
- 849 [7] E. W. Steyerberg, A. J. Vickers, N. R. Cook, T. Gerds, M. Gonen,
850 N. Obuchowski, M. J. Pencina, M. W. Kattan, Assessing the performance
851 of prediction models: a framework for traditional and novel measures,
852 *Epidemiology* 21 (1) (2010) 128–138.
- 853 [8] L. L. Guo, S. R. Pfohl, J. Fries, A. E. W. Johnson, J. Posada,
854 C. Aftandilian, N. Shah, L. Sung, Evaluation of domain general-
855 ization and adaptation on improving model robustness to temporal
856 dataset shift in clinical medicine, *Scientific Reports* 12 (1) (2022) 2726.
857 [doi:10.1038/s41598-022-06484-1](https://doi.org/10.1038/s41598-022-06484-1).
- 858 [9] J. S. Hinson, E. Y. Klein, A. Smith, et al., Multisite implementation
859 of a workflow-integrated machine learning system to optimize COVID-
860 19 hospital admission decisions, *npj Digital Medicine* 5 (1) (2022) 94.
861 [doi:10.1038/s41746-022-00646-1](https://doi.org/10.1038/s41746-022-00646-1).

- [10] M. Buda, A. Maki, M. A. Mazurowski, A systematic study of the class imbalance problem in convolutional neural networks, *Neural Networks* 106 (2018) 249–259.
- [11] A. Alexandari, A. Kundaje, A. Shrikumar, Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation, *International Conference on Machine Learning (ICML)* (2020) 222–232.
- [12] S. E. Davis, R. A. Greevy, C. Fonnesbeck, T. A. Lasko, C. G. Walsh, M. E. Matheny, Detection of calibration drift in clinical prediction models to inform model updating, *Journal of Biomedical Informatics* 112 (2020) 103611.
- [13] G. King, L. Zeng, Logistic regression in rare events data, *Political Analysis* 9 (2) (2001) 137–163.
- [14] A. Caplin, D. Martin, P. Marx, Calibrating for class weights by modeling machine learning, *arXiv preprint arXiv:2205.04613* (2022).
- [15] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, *International Conference on Machine Learning (ICML)* (2017) 1321–1330.
- [16] J. Platt, Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods, *Advances in Large Margin Classifiers* 10 (3) (1999) 61–74.
- [17] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, *Proceedings of the IEEE International Conference on Computer Vision (ICCV)* (2017) 2980–2988.
- [18] M. Piccininni, M. Wechsung, B. Van Calster, J. L. Rohmann, S. Konigorski, M. van Smeden, Understanding random resampling techniques for class imbalance correction and their consequences on calibration and discrimination of clinical risk prediction models, *Journal of Biomedical Informatics* 155 (2024) 104666. [doi:10.1016/j.jbi.2024.104666](https://doi.org/10.1016/j.jbi.2024.104666).
- [19] Y. Geifman, R. El-Yaniv, Selective classification for deep neural networks, *Advances in Neural Information Processing Systems (NeurIPS)* 30 (2017) 4878–4887.

- 893 [20] B. Kompa, J. Snoek, A. L. Beam, Second opinion needed: communicating
894 uncertainty in medical machine learning, *npj Digital Medicine* 4 (1) (2021)
895 4. doi:[10.1038/s41746-020-00367-3](https://doi.org/10.1038/s41746-020-00367-3).
- 896 [21] A. Fisch, T. Schuster, T. Jaakkola, R. Barzilay, Calibrated selective clas-
897 sification, *Advances in Neural Information Processing Systems (NeurIPS)*
898 35 (2022) 28028–28042.
- 899 [22] Y. Aperstein, T. Amit, G. Alon, V. Tal, S. D. Ravit, A. Apartsin,
900 Knowing when not to decide: multi-pathology chest X-ray classification
901 with rejection mechanisms, *Scientific Reports* 16 (2026) 66294. doi:
902 [10.1038/s41598-026-66294-7](https://doi.org/10.1038/s41598-026-66294-7).
- 903 [23] A. E. W. Johnson, T. J. Pollard, S. J. Berkowitz, N. R. Greenbaum, M. P.
904 Lungren, C.-y. Deng, R. G. Mark, S. Horng, MIMIC-CXR, a de-identified
905 publicly available database of chest radiographs with free-text reports,
906 *Scientific Data* 6 (1) (2019) 317. doi:[10.1038/s41597-019-0322-0](https://doi.org/10.1038/s41597-019-0322-0).
- 907 [24] A. Johnson, M. Lungren, Y. Peng, Z. Lu, R. Mark, S. Berkowitz, S. Horng,
908 MIMIC-CXR-JPG - chest radiographs with structured labels, <https://physionet.org/content/mimic-cxr-jpg/2.1.0/> (2024). doi:[10.13026/jsn5-t979](https://doi.org/10.13026/jsn5-t979).
- 911 [25] G. Huang, Z. Liu, L. Van Der Maaten, K. Q. Weinberger, Densely
912 connected convolutional networks, *Proceedings of the IEEE Conference*
913 *on Computer Vision and Pattern Recognition (CVPR)* (2017) 4700–4708.
- 914 [26] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image
915 recognition, *Proceedings of the IEEE Conference on Computer Vision*
916 *and Pattern Recognition (CVPR)* (2016) 770–778.